

ROADMAP · PYTHON REQUIRED · ABOUT 14 WEEKS

Gen AI Roadmap

Six stages from calling a model to shipping an agent that reads your documents, uses tools and is tested before it goes live.

LLMs

RAG

AI Agents

Evals

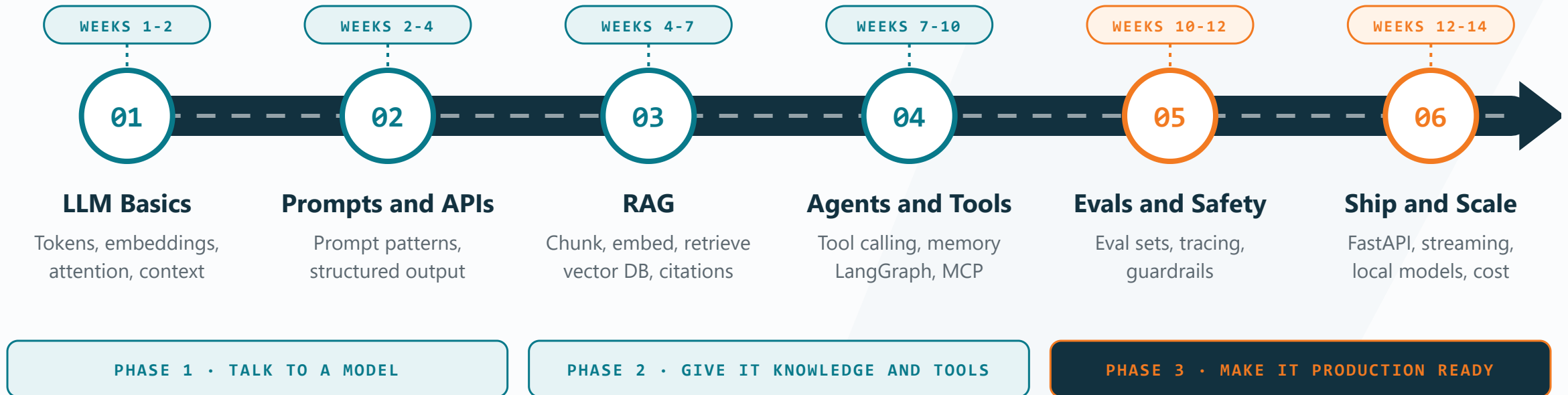
6 Projects



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The Gen AI Roadmap at a Glance

02



You only need **Python and APIs** to start. Learn the concepts, not one model: models change every few months, the stages do not.

Stages 1 to 3: LLMs, Prompts and RAG

STAGE 01 · WEEKS 1-2

LLM Basics

- Tokens, context window and cost
- How transformers and attention work
- Embeddings: meaning as numbers
- Temperature and sampling

[Transformers](#)[Tokenizers](#)

You can: Explain what happens between prompt and reply

STAGE 02 · WEEKS 2-4

Prompts and APIs

- System prompts, roles, few-shot
- Call hosted and local models by API
- Structured output with Pydantic
- Streaming and context engineering

[OpenAI SDK](#)[Ollama](#)[Pydantic](#)

You can: Get a reliable, typed answer from any model

STAGE 03 · WEEKS 4-7

RAG

- Load and chunk PDFs and web pages
- Embed and store in a vector database
- Retrieve, rerank, answer with citations
- Hybrid search and agentic RAG

[Qdrant](#)[Chroma](#)[Docling](#)

You can: Answer questions over your own documents

Stages 4 to 6: Agents to Production

STAGE 04 · WEEKS 7-10

Agents and Tools

- Tool calling and the agent loop
- Short and long term memory
- Graphs and multi-agent workflows
- Connect tools through MCP

LangChain

LangGraph

MCP

You can: Build an agent that plans and acts

STAGE 05 · WEEKS 10-12

Evals and Safety

- Build a test set, score every change
- LLM as a judge, RAG metrics
- Tracing and observability
- Guardrails and prompt injection

LangSmith

Ragas

Langfuse

You can: Prove it works, catch regressions

STAGE 06 · WEEKS 12-14

Ship and Scale

- FastAPI backend and a chat UI
- Caching, latency and token cost
- Serve open models: vLLM, Ollama
- LoRA fine-tuning, only if needed

FastAPI

Docker

vLLM

LoRA

You can: Run a Gen AI app for real users

Projects That Prove Each Stage

STAGE	BUILD THIS	IT PROVES
01 LLM Basics	Token and cost calculator that compares prompts side by side	How models read text
02 Prompts and APIs	Resume parser that returns clean JSON from any PDF	Reliable structured output
03 RAG	Chat with your PDFs, every answer with page citations	Grounding and retrieval
04 Agents and Tools	Research agent that searches, reads and writes a report	Tool use and planning
05 Evals and Safety	Eval suite that scores your RAG app on 50 questions	Measuring quality
06 Ship and Scale	Your RAG agent deployed with a UI, streaming and tracing	End to end ownership

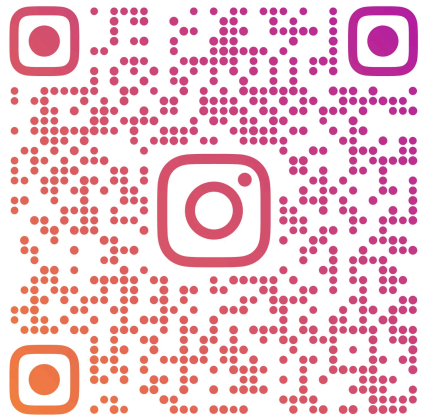
Capstone: join stages 3 to 6 into one app. A document assistant with tools, evals and a public URL beats ten tutorials.

GEN AI ROADMAP · SIX STAGES

Thank You

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